MIL-L-741D
10 August 1984
SUPERSEDING
MIL-L-741C
7 October 1977

MILITARY SPECIFICATION

LEADS, ELECTRICAL, ARC-WELDING

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers arc-welding electrical, single-conductor leads, with heavy-duty jacket.

1.2 Classification. Arc-welding electrical leads shall be of the following sizes and grades, as specified (see 6.2):

Size 2 - No. 2 American Wire Gage (AWG).
Size 1/0 - No. 1/0 AWG.
Size 2/0 - No. 2/0 AWG.

Grade A - Medium-low temperature.
Grade B - Low temperature.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6150

Source: https://assist.dla.mil -- Downloaded: 202610-16T11:23Z
Check the source to verify that this is the current version before use.

PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-B-640 - Boxes, Fiberboard, Corrugated, Flexible.

MILITARY

MIL-P-116 - Preservation, Methods of.
MIL-I-3930 - Insulating and Jacketing Compounds, Electrical (for cables, cords, and wires).
MIL-T-7928 - Terminals, Lug: Splices, Conductor: Crimp Style, Copper, General Specification For.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-147 - Palletized Unit Loads.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D3951 - Commercial Packaging, Standard Practice for.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

NEMA WC3 - Rubber-Insulated Wire and Cable for the Transmission and Distribution of Electrical Energy.

(Application for copies should be addressed to the National Electrical Manufacturers Association, 2101 L Street, N.W., Washington, D.C. 20037.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T11:23Z
Check the source to verify that this is the current version before use.

3. REQUIREMENTS

3.1 Description. The leads shall consist of lengths of rope lay bunch-stranded copper wire covered with separator and thermosetting jacket and terminated at each end with a copper lug.

3.2 Standard commercial product. The leads, electrical, acr-welding shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer's standard commercial product. Additional or better features which are not specifically prohibited by this specification but which are a part of the manufacturer's standard commercial product, shall be included in the leads being furnished. A standard commercial product is a product which has been sold or is being currently offered for sale on the commercial market through advertisements or manufacturer's catalogs, or brochures, and represents the latest production model.

3.3 First article. When specified (see 6.2), the contractor shall furnish a lead of each size and grade for first article inspection and approval (see 4.2.1 and 6.3).

3.4 Material. All material shall be as specified herein. Material not specified shall be of the same quality used for the intended purpose in commercial practice. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual component or the overall assembly.

3.5 Construction. The leads shall conform to NEMA WC-3, as applicable, for portable arc-welding cables. Cable for leads shall have been manufactured not more than 6 months prior to the date of acceptance of the leads. Unless otherwise specified (see 6.2), the lead shall be 50 feet in length and terminated at each end with a one-hole, crimp type, copper lug (see 3.8).

3.6 Conductor. The conductor shall be rope-lay, bunch-stranded, No. 34 AWG copper wire. The separator shall be either laminated metallic foil, heavy paper, textile serving polyester tape or varnished cambric.

3.7 Jacket. Unless otherwise specified (see 6.2), the jacket shall be made of one of the following compounds: heavy-duty chloroprene rubber (neoprene), ethylene-propylene-diene rubber, heavy-duty vulcanized acrylonitrile-butadiene/polyvinyl-chloride compound, chlorinated polyethylene or heavy-duty vulcanized chlorosulfonated-polyethylene compound. EPDM rubber shall not be used as a jacket which is required to be oil resistant and degreaser solvent resistant as specified (see 6.2).

3.7.1 Grade A leads. When specified (see 6.2), the jacket used for grade A leads (see 1.2) shall, in addition to meeting the requirements specified in 3.6, conform to MIL-I-3930, type JN, medium-low temperature jacketing compounds.

3.7.2 Grade B leads. The jacket used for grade B leads (see 1.2) shall, in addition to meeting the requirements specified in 3.6, conform to MIL-I-3930, type JN-L or type JH requirements for low-temperature jacketing compounds.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T11:23Z
Check the source to verify that this is the current version before use.

3.8 Terminal lugs. Terminal lugs shall conform to MIL-T-7928 for one-hole, crimp type, copper lugs, and shall be firmly affixed to the ends of the leads with a suitable crimping tool. The lug joints of complete leads shall withstand a pull of 400 pounds for 1 minute when tested as specified in 4.6.2. The millivolt drop through the crimped joint shall not exceed 8 millivolts for any size leads, when tested as specified in 4.6.3.

3.9 Identification marking. The lead shall be provided with a durable marking on the jacket throughout its entire length, at intervals not exceeding 2 feet, by which it may be identified as the product of a particular manufacturer. It shall include the size number AWG, grade of jacket, and the date of manufacture of the lead. The marking shall consist of lettering, not code. The manufacturer's code and part number may also be shown.

3.10 Workmanship. The quality of workmanship shall be such as to produce leads that are in accordance with the requirements of this specification and are so fabricated as to insure proper functioning of the unit.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements specified herein and in applicable referenced documents.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.2.1).
(b) Quality conformance inspection (see 4.2.2).

4.2.1 First article inspection. First article inspection shall be performed on one lead of each size and grade when a first article is required (see 3.2 and 6.2). This inspection shall include the examination of 4.4 and the tests of 4.5. The first article may be a standard production item from the supplier's current inventory provided the lead meets the requirements of the specification and is representative of the design, construction, and manufacturing technique applicable to the remaining leads to be furnished under the contract.

4.2.2 Quality conformance inspection. The quality conformance inspection shall include the examination of 4.4, the tests of 4.5, and the packaging inspection of 4.6. This inspection shall be performed on the samples selected in accordance with 4.3.

Source: https://assist.dla.mil -- Downloaded: 202640-16T11:23Z
Check the source to verify that this is the current version before use.

4.3 Sampling. Sampling and inspection procedures shall be in accordance with MIL-STD-105. All leads offered for delivery at one time shall be considered a lot for the purpose of inspection. If an inspection lot is rejected, the contractor may rework it to correct the defects, or screen out the defective units, and resubmit for a complete reinspection. Resubmitted lots shall be reinspected using tightened inspection. If the rejected lot was screened, reinspection shall be limited to the defect causing rejection. If the lot was reprocessed, reinspection shall be performed for all defects. Rejected lots shall be separate from new lots, and shall be clearly identified as reinspected lots.

4.3.1 Sampling for examination. Examination shall be based on inspection level II and an Acceptable Quality Level (AQL) of 4.0 percent defective.

4.3.2 Sampling for tests. Tests shall be based on inspection level S-4 and an AQL of 6.5 percent defective.

4.4 Examination. Each lead shall be examined for compliance with the requirements specified in section 3 of this specification. Any redesign or modification of the contractor's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirements or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

4.5 Tests. The first article sample, when a first article sample is required, and each sample selected in accordance with 4.3.2 shall be tested in accordance with NEMA WC-3 and MIL-I-3930, as applicable. Tests shall include, but shall not necessarily be limited to, verification of the properties and characteristics as specified when tested as specified in 4.5.1, 4.5.2, and 4.5.3. Failure to pass any phase of the tests shall be cause for rejection.

4.5.1 Physical and aging tests. Physical and aging tests shall be conducted on the selected samples of each grade of jackets to determine conformance to 3.7.

4.5.2 Tensile strength. The tensile strength and security of terminal lugs shall be determined by means of a suitable testing machine or a free-hanging weight. Lug terminals and conductor shall not break or separate when the specified force (see 3.8) is applied.

4.5.3 Millivolt drop test. The millivolt drop shall be measured when the cable is loaded to the following current values:

AWG size	Current (amperes)
2	200
AWG size	Current (amperes)
1/0	300
2/0	375

Source: https://assist.dla.mil -- Downloaded: 202640-16T11:23Z
Check the source to verify that this is the current version before use.

Millivolt drop measurements shall be made by puncturing the insulation of the conductor 1/16-inch back of cable receiving end of the lug and at the intersection of barrel and tongue of the lug for the test points. The millivolt drop between the test points shall not exceed the value specified herein (see 3.8).

4.6 Packaging inspection. The preservation, packing, and marking of the item shall be inspected to verify conformance to the requirements of section 5.

5. PACKAGING

5.1 Preservation. Preservation shall be level A, B, or C as specified (see 6.2).

5.1.1 Level A. Each lead shall be preserved method IC 2 in accordance with MIL-P-116. Each lead shall be coiled to the minimum safe diameter and the coil shall be secured. Each coil shall be placed in a box conforming to PPP-B-636, class domestic before being given method IC 2 preservation.

5.1.2 Level B. Preservation shall be the same as level A except that method III shall be used with a box conforming to PPP-B-636, class weather-resistant and closed method V in accordance with the appendix of PPP-B-636.

5.1.3 Level C. Material shall be preserved and packaged in accordance with ASTM D3951.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A. The leads shall be packed in containers conforming to PPP-B-601, overseas type or PPP-B-621, class 2. The contents shall be cushioned, blocked and braced to prevent movement or damage.

5.2.2 Level B. The leads shall be packed in containers conforming to PPP-B-601, domestic type; PPP-B-621, class 1; PPP-B-636, class weather-resistant, or PPP-B-640, class 2. The contents shall be cushioned, blocked and braced to prevent movement or damage.

5.2.3 Level C. Material shall be packed in accordance with ASTM D-3951.

5.3 Palletization. Material shall be palletized in accordance with MIL-STD-147 when the following criteria are met:

a. Load to consist of four or more unskidded containers; and,
b. Load shall utilize a minimum of 80 percent of the pallet base.

5.4 Marking. Marking shall be in accordance with MIL-STD-129.

6. NOTES.

6.1 Intended use. The leads covered by this specification are intended for use between arc-welding electrode holders or trailing cables and power outlets.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T11:23Z
Check the source to verify that this is the current version before use.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Size and grade of lead required (see 1.2).
c. When a first article is required for inspection and approval (see 3.3, 4.2.1, and 6.3).
d. Length required if other than specified (see 3.5).
e. When the jacket shall be oil and solvent resistant (see 3.7).
f. When the jacket material for grade A leads shall conform to MIL-T-3930, type JN (see 3.7.1).
g. Level of preservation and level of packing required (see 5.1 and 5.2).

6.2.1 Contract data requirements. When this specification is used in a procurement which incorporates a DD Form 1423 and invokes the provisions of paragraph 52.227-7031 of the Federal Acquisition Regulation (FAR). The data requirements will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the Contract Data Requirements Lists (DD Form 1423) incorporated into the contract. When the provisions of FAR 52.227-7031 are not invoked, the data shall be delivered in accordance with the contract requirements.

6.3 First article. When a first article is required, it shall be tested and approved under the appropriate provisions of paragraph 52.209-3 of the FAR. The first article may be a standard production item from the contractor's current inventory as specified in 4.2.1. The first article should consist of one lead of each size and grade. The contracting officer should include specific instructions in all procurement instruments, regarding arrangements for examinations, test and approval of the first article.

6.4 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:

Army - ME
Navy - YD
Air Force - 99

Preparing Activity:

Navy - YD

(Project 6150-0154)

Review Activities:

Army - CR, ER
DLA - GS
Air Force - 80

Source: https://assist.dla.mil -- Downloaded: 202640-16T11:23Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T11:23Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE NAVY
Commander (04M2)
Naval Facilities Engineering Command
200 Stovall Street
Alexandria, VA 22332-2300

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO 12503 WASHINGTON D C

POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE NAVY

Commanding Officer (1561M)
Naval Construction Battalion Center
Port Hueneme, CA 93043-5000

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T11:23Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426

PREVIOUS EDITION IS OBSOLETE.

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T11:23Z
Check the source to verify that this is the current version before use.